Vidhata Jayaraman

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Education

University of Illinois Urbana-Champaign (UIUC) — Anticipated Graduation: May 2026
B.S. in Computer Engineering | B.S. in Mathematics — *GPA: 3.97/4.0*
Dean's List (Top 20% of Student Body)

Research Experience

Research with Professor Lav Varshney at UIUC — February 2024 – Present

Current Projects

- **Senior Thesis**: *Information-theoretic lower bound for Knowledge Distillation in LLMs* (Primary researcher): Attempting to find information-theoretic lower bounds for knowledge distillation with a focus on LLMs
- *Extending Community Detection/Graph Clustering analysis to Quantum Networks* (Primary researcher): Extending the "No Free Lunch" Theorem and community detection algorithms from the classical graph setting to quantum networks

Past Projects

- *Emergent Capabilities in Transformers*: Experimenting with Modern Hopfield Networks and other Neural Associative Memories to understand emergent capabilities as model size increases and their connection to Transformers
- *SwitchCIT: Switching for Continual Instruction Tuning of Large Language Models*: Identified clustering in the final layer of an LLM following continual learning and used this to create a switch network which helps avoid catastrophic forgetting

Research with Professor Xu Chen at UIUC — March 2023 – February 2024

- Created a Physics Informed Neural Network (PINN) to model Voltage and Electric field from a charged circle/sphere inside a grounded box.
- Utilized a version of the Deep Galerkin Method (DGM) to estimate the differential equation modeling the system.
- Implementation was used & modified by Samsung engineers for use in their own research and modeling.
- Implemented an operator estimator for an RLC (Resistor, Inductor, Capacitor) circuit to model for any R, L, and C.

Industry Experience

AbbVie | *Machine Learning Research Intern* — May 2025 – August 2025

- Implemented the SubgraphX method for GNN explainability, modified for multi-target regression and use with 3D input.
- Method correctly identified many important regions in molecules for property prediction, strengthening an internal site-of-metabolism tool.
- Conducted novel Uncertainty Quantification research using metric learning and distribution parameter estimation.

Johns Hopkins University Applied Physics Laboratory | *Data Science Intern* — June 2024 – August 2024

- Created a chat-bot with Retrieval Augmented Generation (RAG) for retrieving information in technical documents.
- Implemented a Bayesian approach towards reliability growth planning (RGP) using advanced optimization techniques.
- Manuscript in preparation to be submitted to IEEE.

National Institute of Standards and Technology (NIST) | *Research Intern* — June 2023 – August 2023

- Created a small GPT-2 model which utilized Cloze Probabilities to identify abnormal sentences in text data.
- Demonstrated that analysis of outliers in dimensionally reduced text embeddings can provide similar results, and compared dimensionality reduction methods.

Brunswick i-JET Research Lab | *Autonomous Simulation Intern* — January 2023 – May 2023

- Utilized Robotic Operating System (ROS) to create maps using Simultaneous Localization and Mapping (SLAM).
- Utilized theories of fluid dynamics and wake physics to build an autonomous "perfect" wake generator and used ROS to visualize and analyze data.

Teaching and Mentorship

Algorithms & Models of Computation Classroom Assistant — August 2025 – Present

- Assisting in quiz and exam grading and holding 2-hour-long office hours weekly.
- Writing lecture notes on the connection between the WL graph isomorphism test and the expressivity of Graph Neural Networks.

Analog Signal Processing (ECE 210) Tutor (IEEE-HKN) — August 2023 – May 2024

- Provided 1-on-1 tutoring, created review slideshows/worksheets, and gave review lectures before midterms.

Skills & Expertise

Deep Learning: PyTorch, TensorFlow
Natural Language: NLTK, spaCy, LangChain
Software: Python, C/C++, Java
Web/API: Flask, Django, HTML/CSS, Javascript
DevOps: Git, Docker
Low-Level: x86 assembly, SystemVerilog
Robotics: ROS, OpenCV
Research/Math: LaTeX

Research Interests

Machine Learning & Optimization; Information Theory & Quantum Information; High-dimensional Statistics; Geometric Analysis & Uncertainty Quantification

Relevant Coursework

Math: Real Analysis; Complex Analysis; Abstract Algebra; Linear Algebra; Probability Theory; Algebraic Topology; Graph Theory; Differential Equations; Probability & Measure

Applied Math/CS: Optimization; Machine Learning; Deep Learning for CV; Deep Generative Models; Random Processes; Information Theory; Quantum Information Theory; Algorithms; Data Structures; Signal Processing

Grants & Awards

- Grant for undergraduate research from the Office of Undergraduate Research 2025
- UIUC James Scholar 2023 – 2025
- Inducted into Eta Kappa Nu (IEEE-HKN), an ECE Honors Society 2023
- Inducted into Tau Beta Pi, the Engineering Honor Society 2023
- Selected for American Invitational Mathematics Examination (AIME) 2022
- Illinois State Seal of Biliteracy in Spanish 2020

Publications & Conferences

1. Hartman, M., **Jayaraman V.A.**, Choaria, M., Bhimaraju, A., & Varshney, L.R. (2025). Skip-It? Theoretical Conditions for Layer Skipping in Vision-Language Models. Preprint arXiv:2509.25584 *(Under review for ICLR 2026)*
2. Accepted to the first AI Startup School Conference (June 2025) hosted by YCombinator for accomplished undergraduate and graduate researchers.
3. Hartman, M., **Jayaraman, V.A.**, , Choraria, M., Bhimaraju, A., & Varshney, L. R. (2025). Unmasking the functionality of early layers in VLMs *(To be presented at eXCV Workshop @ ICCV 2025)*
4. Wu, X., Hartman, M., **Jayaraman, V.A.**, & Varshney, L. R. (2024). SwitchCIT: Switching for Continual Instruction Tuning of Large Language Models. Preprint arXiv:2407.11780. *(Under review for JSTSP 2025)*
5. Bernstein, H.C., Bindel, S.R., McKibben, M.A., & **Jayaraman, V.A.** (2024). Planning Model based on Projection Methodology Bayesian Discrete Extended (PM2-BDE) *(Under internal review)*

Projects

RAG Chat-bot for ECE 391 (Computer Systems Engineering)
Python, LangChain, dash July 2024 – September 2024

- Created a RAG chat-bot to search through ECE 391 project documents, citing the document title and page number.
- Implemented cutting-edge RAG techniques such as parent-child text splitting and cross-encoder reranking.

Operating System
C, x86 assembly March 2024 – May 2024

- Developed kernel for a Linux-based OS with interrupt-based device I/O support.
- Implemented radix-2 paging, Round-Robin scheduling, and a file system capable of 4MB files.
- Created a Linux shell for user command input of basic Linux shell commands.

Named Entity Highlighter | Chrome Extension
Javascript, Python, PyTorch, Django July 2023 – August 2023

- Trained and implemented a state-of-the-art Named Entity Recognition (NER) AI model from scratch.
- Developed a Chrome Extension to highlight and provide Wikipedia links to named entities on a web page.